

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Publication

- “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages

Python, C++, CUDA, SQL, Bash

Vision-Language Models

VLM architecture design, contrastive image-text pretraining (CLIP-style), visual instruction tuning (SFT), preference optimization (RLHF / DPO / GRPO), LLaVA / BLIP / Qwen-VL-style models

Generative age/Video/3D)

(Im- latent diffusion, DiT, ControlNet conditioning, text-to-image/video generation & editing, NeRF / Gaussian Splatting, 3D-aware synthesis

Efficient Multimodal

knowledge distillation, mixture-of-experts (MoE), quantization, LoRA / QLoRA, pruning, on-device inference

Frameworks & Infra

PyTorch, HuggingFace Transformers / Diffusers, JAX, DeepSpeed / FSDP, distributed multi-GPU training, Weights & Biases, AWS, GCP, Azure

Research & Projects

Vision-Language Model for Multimodal Understanding & Reasoning

PyTorch, HuggingFace, DeepSpeed

VLM pretraining + post-training | image/video/text

- Designed and pretrained a **vision-language model** with a CLIP-style image encoder aligned to an LLM via contrastive image-text pretraining and a learned projector, scaling multi-GPU training with DeepSpeed/FSDP.
- Applied **post-training** (visual instruction tuning via SFT, then preference optimization with DPO/GRPO) to improve multimodal **retrieval, VQA, and reasoning**, evaluating on standard image and video QA benchmarks.

Diffusion-Based Image / Video Generation & Editing

PyTorch, Diffusers

latent diffusion, DiT, ControlNet | text-guided generation & editing

- Built a **latent diffusion** pipeline with a DiT backbone for text-to-image **generation and editing**, adding ControlNet-style conditioning for structure-preserving edits and temporal layers for short-video synthesis.
- Extended the approach toward **3D-aware generation** (NeRF / Gaussian-Splatting representations) and benchmarked sample quality (FID/CLIP-score) against strong baselines.

Efficient Multimodal Models via Distillation & Mixture-of-Experts

PyTorch, CUDA

knowledge distillation, MoE, on-device inference

- Compressed a large VLM into a compact student via **knowledge distillation** and a **mixture-of-experts** router, cutting inference cost while retaining most accuracy for on-device deployment.
- Applied quantization and LoRA-based adaptation to fit the model within tight memory budgets, profiling latency and throughput trade-offs across hardware.

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models; replaced spreadsheet workflows and accelerated iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.